

Synopsis

Disease Risk Prediction using Machine Learning Classification

Problem Id: 01

Student Name-

- Arnav Kotnala (2028397)
- Ayushi Rautela (2028406)
- Siddhant Singh (2028498)

Section- L

Problem Description:

The prediction of disease risk through analysis of patient clinical data establishes the foundation for early diagnosis and preventive medical treatment. The project develops a complete machine learning classification system which uses patient health data to forecast disease risk.

The system starts with the selection of a clinical dataset which undergoes data preprocessing that includes missing value handling and data normalization and data transformation to create high-quality input materials. We employ feature selection methods to determine which medical attributes most significantly impact our analysis. The results improve model performance while making it easier to understand how the models function.

The research tests different classification techniques by using Logistic Regression and KNearest Neighbors (KNN) and Support Vector Machine (SVM) methods. The model evaluation uses standard classification indicators which include accuracy and precision and recall and F1-score and confusion matrix.

The project demonstrates a complete machine learning process which transforms raw clinical data into validated predictive models while implementing best machine learning system development practices.

Steps of Implementation:

1. Dataset Curation (Data Collection)

The dataset used in this project contains 500 records and 11 attributes related to patient clinical and lifestyle information. Each row represents an individual patient, and each column represents a specific feature used to analyze and predict disease risk.

The dataset includes the following attributes:

- Patient_ID: Unique identifier for each patient
- Age: Age of the patient (numerical)
- Gender: Gender of the patient (Male/Female)
- Blood_Pressure: Blood pressure level (numerical)
- Cholesterol: Cholesterol level (numerical)
- Glucose: Blood glucose level (numerical)
- BMI: Body Mass Index (numerical)
- Smoking: Indicates if the patient smokes (Yes/No)
- Alcohol: Indicates alcohol consumption (Yes/No)
- Physical_Activity: Level of physical activity (Low/Medium/High)
- Disease_Risk: Target variable indicating disease risk (High/Medium/Low or Yes/No)

The dataset consists of both numerical and categorical features. Some missing values and inconsistencies may be present, which require preprocessing such as data cleaning, handling null values, and normalization.

This dataset is used to build a machine learning classification model to predict whether a patient is at risk of disease based on the given clinical parameters.

2. Data Cleaning (Preprocessing)

Here, cleaning of the data set was done to make it better before building the model. The missing values were taken care of with the help of mean/median so that all records could be considered while building the model. Duplicates were eliminated to prevent the repetition of the same records.

The outliers were also treated with great care since the presence of outliers may have an impact on the performance of the machine learning models. The values were kept consistent throughout the entire data set so that the consistency of the data is maintained. Moreover, the data set was scanned for any inconsistencies. Thus, the data set became more reliable after performing this preprocessing task. Steps Involved:

- Handle missing values using mean/median imputation Missing values were filled using mean or median values.
This helped make the dataset complete and usable.
- Remove duplicate records
Duplicate rows were identified and removed from the dataset.
This improved data accuracy and reduced repetition.
- Detect and handle outliers
Outliers were checked in numerical columns.
They were handled to reduce their effect on the model.
- Ensure consistency in data formats
Data values were converted into a consistent format.
This made the dataset clean and easier to analyze.

3. Feature Standardization (StandardScaler)

The team uses StandardScaler to normalize data because clinical features contain different measurement systems and International System of Units. The process converts all features into a standardized format which results in a mean value of 0 and a standard deviation of 1. The current step shows its significance because K-Nearest Neighbours (KNN) and Support Vector Machine (SVM) algorithms require proper feature scaling for their operations. Proper standardization improves model performance, convergence speed, and prediction accuracy.

4. Model Building (Classification Models)

Classification models are trained using the processed clinical dataset to predict disease risk. The models used in this project include Logistic Regression, KNN, and SVM.

These models learn the relationship between input features such as age, BMI, blood pressure, glucose level, and insulin levels and the target variable (disease risk).

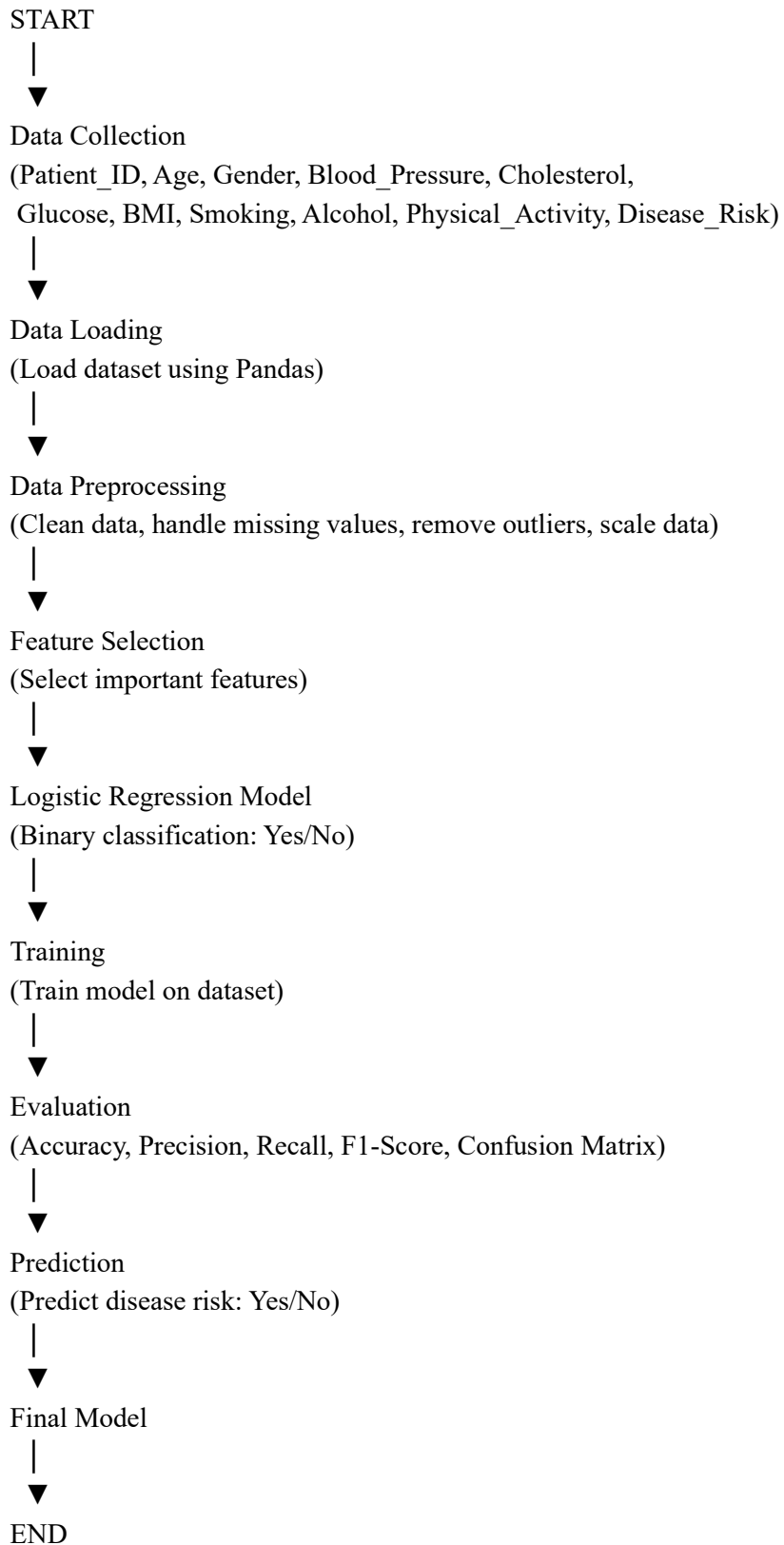
5. Model Evaluation(Performance Analysis)

The trained models are evaluated using standard classification metrics to measure their effectiveness. These include accuracy, precision, recall, and F1-score to assess prediction quality.

A confusion matrix is used to visualize correct and incorrect classifications.

These evaluation techniques help identify the best-performing model and highlight areas for improvement.

FLOWCHART



Expected Modules/ Libraries to be used :

- **NumPy:**

Used for numerical operations and efficient handling of arrays.

- **Pandas:**

Used for data manipulation, cleaning, and handling structured datasets.

- **Matplotlib / Seaborn:**

Used for data visualization such as confusion matrix, ROC curves, and performance comparison graphs.

- **Scikit-learn:**

Provides machine learning tools including:

- StandardScaler (for normalization)
- Logistic Regression
- K-Nearest Neighbours (KNN)
- Support Vector Machine (SVM)
- Train-Test Split
- Evaluation metrics

Accuracy represents the proportion of correctly classified

Expected Evaluation metrics and Plots to be used:

1. Accuracy

instances out of the total dataset. It provides a general measure of model performance across all classes.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

While accuracy is easy to interpret, it can be misleading in cases of class imbalance. For example, in a disease prediction dataset where healthy patients dominate, a model predicting all cases as “healthy” may still achieve high accuracy without being clinically useful.

2. Precision

Precision measures the proportion of correctly predicted positive observations out of all predicted positives.

$$Precision = \frac{TP}{TP + FP}$$

It answers the question: “When the model predicts a patient has the disease, how often is it correct?”

High precision is crucial in scenarios where false positives are costly, such as unnecessary treatments or anxiety for patients.

3. Recall

Recall, also known as sensitivity or true positive rate, measures the proportion of actual positives correctly identified by the model.

$$Recall = \frac{TP}{TP + FN}$$

It answers: “Out of all actual disease cases, how many did the model correctly detect?”

In healthcare applications, recall is often prioritized because missing a true disease case (false negative) can have serious consequences.

4. F1-Score

The F1-score is the harmonic mean of precision and recall, providing a balance between the two.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

It is particularly useful when dealing with imbalanced datasets, where optimizing only precision or recall may lead to biased models. The harmonic mean ensures that both metrics are reasonably high; if one is low, the F1-score drops significantly.

5. Confusion Matrix

The confusion matrix is a tabular representation that summarizes the classification performance of a model. It consists of four components:

- True Positives (TP): Correctly predicted disease cases
- True Negatives (TN): Correctly predicted healthy cases
- False Positives (FP): Healthy patients incorrectly predicted as diseased
- False Negatives (FN): Diseased patients incorrectly predicted as healthy

This matrix provides deeper insight into types of errors made by the model rather than just overall accuracy. It is especially important in clinical systems where different errors have different implications.

A confusion matrix also serves as the foundation for calculating other evaluation metrics such as precision, recall, and F1-score.

6. ROC Curve (Receiver Operating Characteristic Curve)

The ROC Curve is a graphical representation used to evaluate the performance of a classification model at different threshold values. It plots the True Positive Rate (TPR) against the False Positive Rate (FPR).

$$TPR(Recall) = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

The ROC curve helps in understanding how well the model distinguishes between positive and negative classes.

Required topics from the subject:

- Matplotlib
- Seaborn
- Scikit-learn (sklearn)
- Exploratory Data Analysis (EDA)
- Data Preprocessing
- Feature Selection (Forward Selection & Backward Elimination)
- Normalization Techniques (Min-Max Scaling & Standard Scaling)
- Logistic Regression
- K-Nearest Neighbors (KNN)
- Evaluation Metrics (Accuracy, Precision, Recall, F1-Score, Confusion Matrix)

Platform Required:

Jupyter Notebook:

Used as the primary environment for coding, testing, and visualization.

Python (Anaconda Distribution):

Provides all required libraries and package management needed for data analysis and machine learning.

Books and Link Sources:

- <https://scikit-learn.org/stable/>
- <https://www.geeksforgeeks.org/machine-learning/>
- <https://www.geeksforgeeks.org/logistic-regression-in-machine-learning/>
- <https://www.geeksforgeeks.org/k-nearest-neighbours/>

Data Set-

https://github.com/arnavkotnala/AIML_PBL_TEAM_ID-1/blob/main/finalizedd.xlsx